

# Saurabh Koju

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## EDUCATION

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- **Bachelors in Computer Engineering** November 2019 – March 2024  
*Pulchowk Campus, Institute of Engineering* Lalitpur, Nepal
  - Rank 54 / 12,000 (**top 0.5%**) in national entrance exam.
  - Graduated in *First Division* with **78.97%**
- **Certifications**
  - **GRE:** 337 (Quant 170, Verbal 167)
  - **IELTS:** 8.5 (Listening 9.0, Reading 9.0, Writing 8.0, Speaking 7.5)

## RESEARCH EXPERIENCE

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- **Research Assistant** October 2024 - August 2026  
*Multimodal Learning Lab – Supervisor: [Dr. Binod Bhattarai](#)* University of Aberdeen, UK & NAAMII
  - Working on research projects on **Surgical World Models**, **Medical Imaging** and **Generative Modeling**.
  - Introduced **surgical world model** based on inverse dynamics model that **learns latent actions from unlabeled surgical videos**.
  - Working on a novel method to obtain **latent actions that can be transferred across surgical contexts** with coarse latent action modeling.
  - Contributed to spatial proteomics image generation project with diffusion models.
- **Research Intern** July 2024 - September 2024  
*Multimodal Learning Lab – Supervisor: [Dr. Binod Bhattarai](#)* University of Aberdeen, UK & NAAMII
  - **SAGES CVS Grand Challenge 2024: CVS Criteria Prediction in Cholecystectomy Videos, MICCAI 2024**  
Developed a novel object-token transformer with relative layout embeddings that secured **Fourth position** in two sub-challenges: CVS Classification and Robustness.

## PREPRINTS AND PUBLICATIONS

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- **SurgTransfer: Multi-Step Latent Action Modeling in Surgical World Model.** *Under Review*  
**Saurabh Koju**, Love Panta, Deepak Thapa, Rudra P. K. Poudel, Prajwal Ghimire, Angus Watson, Yash Raj Shrestha, Prashanna Kumar Gyawali & Binod Bhattarai
- **The SAGES Critical View of Safety Challenge: A Global Benchmark for AI-Assisted Surgical Quality Assessment** *Under Review. [Paper](#)*  
Deepak Alapatt, Jennifer Eckhoff, Zhiliang Lyu, ..., **Saurabh Koju**, ..., Nicolas Padoy
- **Surgical Vision World Model.** *Workshop on Data Engineering in Medical Imaging, MICCAI 2025. [Paper](#)*  
**Saurabh Koju**, Saurav Bastola, Prashant Shrestha, Sanskar Amgain, Yash Raj Shrestha, Rudra P. K. Poudel & Binod Bhattarai

## AWARDS AND ACHIEVEMENTS

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- **International Math Olympiad**  
Secured **1st position** in **National Math Olympiad** and represented Nepal at IMO 2019 (Bath, UK)
- **Other Olympiads**
  - **Winner of Various National Math and Physics Competitions:** Physics–Math Olympiad 2019 by Mechatrix, Maths Hunt 2018 by St. Xavier’s College
  - **Top 6** in National Physics Olympiad 2019 pre-selection test
- **Competitive Programming**
  - Achieved **Global Ranking of 76 (1st Nationally)** Among 7091 Teams in IEEEExtreme Programming Competition 17.0

- **Winner of Various National Coding Competitions:** LOCUS Codejam 2022, Yomari Coding Competition 2023, LOCUS Codejam 2023
- Achieved Maximum Rating of **1793** on Codeforces
- **SmartBots** First Runner up at SmartBots Coding Challenge  
*Involved development of an efficient game playing bot for a card game, competition involved 94 teams nationwide*
- **Scholarship** Merit-based full tuition waiver in undergrad based on entrance exam ranking

## TEACHING EXPERIENCES

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- **Teaching Assistant, *Sixth Annual Nepal AI School (AN AIS) 2025*** December 2025  
Delivered lab sessions on geometric deep learning
- **Instructor, *Competitive Programming Workshop, IEEE Pulchowk*** July 2023  
Prepared and delivered a 5-day workshop on competitive programming
- **Instructor, *Software Fellowship, LOCUS 2023*** June 2023  
Prepared and delivered lectures on Data Structure and Algorithms
- **Instructor, *Competitive Programming Workshop, Thapathali Campus*** July 2022  
Prepared and delivered a 6-day workshop on competitive programming

## VOLUNTEER EXPERIENCES

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- **Reviewer, *Workshop on Data Engineering in Medical Imaging, MICCAI 2026*** 2026
- **Reviewer, *Workshop on Data Engineering in Medical Imaging, MICCAI 2025*** 2025
- **Mathematics Coordinator, *Maths and Physics Society, Pulchowk Campus*** 2022 – 2023

## SELECTED ACADEMIC PROJECTS

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- **Automata Toolkit:** Implemented functionalities to convert between DFAs, NFAs and regular expressions, minimize DFAs, and visually simulate DFAs and NFAs.
- **Interpreter for Custom Programming Language:** Implemented a general CFG parser using Earley Parsing; Full fledged implementation of functionalities typical of a high level programming language.
- **VHDL Simulator:** Implemented a subset of VHDL to simulate digital circuits programmatically; Added automatic circuit diagram generation from VHDL specifications.

## TECHNICAL SKILLS

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- **Languages:** Python, C/C++, JavaScript
- **Frameworks:** PyTorch, NumPy, Matplotlib, Scikit-learn, MMDetection
- **Machine Learning:** Computer Vision, Generative Modeling, Reinforcement Learning
- **Other Tools:** Git, LaTeX, Shell Scripting, Docker, Slurm, Linux Environment